Topic 3: Normal model with conjugate prior

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Textbooks

- A. Johnson, M. Ott, and M. Dogucu. (2022), *Bayes Rules!*, CRC Press.
  - Chapter 5: normal-normal Bayesian model
- J. Albert (2007), *Bayesian Computation with R*, Springer Press.
  - Chapter 3: normal model with known mean and unknown variance
  - Chapter 4: normal model with both parameters unknown

Outline

Prior: normal distribution

Gaussian likelihood

Maximum likelihood estimation

Posterior distribution: unknown mean and known variance

Bias variance trade-off

Prior: normal inverse gamma distribution

Posterior distribution: unknown mean and variance

Diffuse priors

Highest posterior density intervals

Marginal likelihood

Posterior predictive distribution

Normal distribution

Definition

A random variable x has a **normal distribution** written $x \sim \mathrm{N}\left(\mu, \sigma^2\right)$, if the probability density function (p.d.f.) is

$$p\left(x|\mu, \sigma^2\right) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) \exp\left(-\frac{1}{2\sigma^2}\left[x-\mu\right]^2\right)$$

where μ is the mean and σ^2 is the variance.

- $\mathbb{E}\left[x\right] = \mu$.
- $\mathbb{V}\left[x\right] = \sigma^2$.
- Kernel: $k\left(x\right) = \exp\left(-\frac{1}{2\sigma^2}\left[x-\mu\right]^2\right)$.
- Normalizing constant: $\frac{1}{\sqrt{2\pi\sigma^2}}$

R code

- Probability density function

```
dnorm(x, mean = 0, sd = 1, log = FALSE)
```

- Generate random numbers

```
rnorm(n, mean = 0, sd = 1)
```

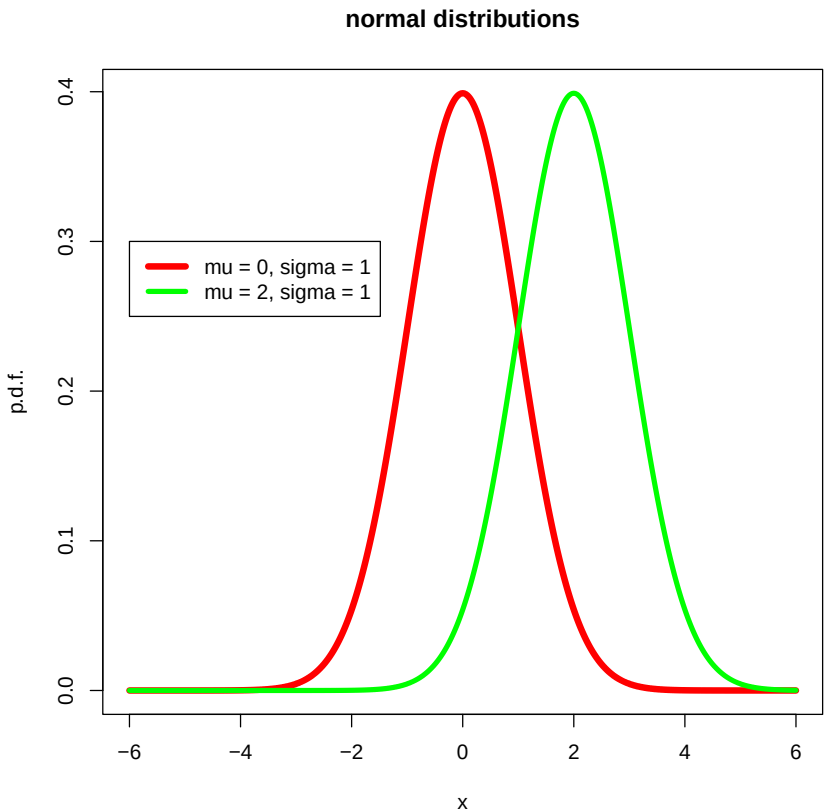
- Cumulative distribution function

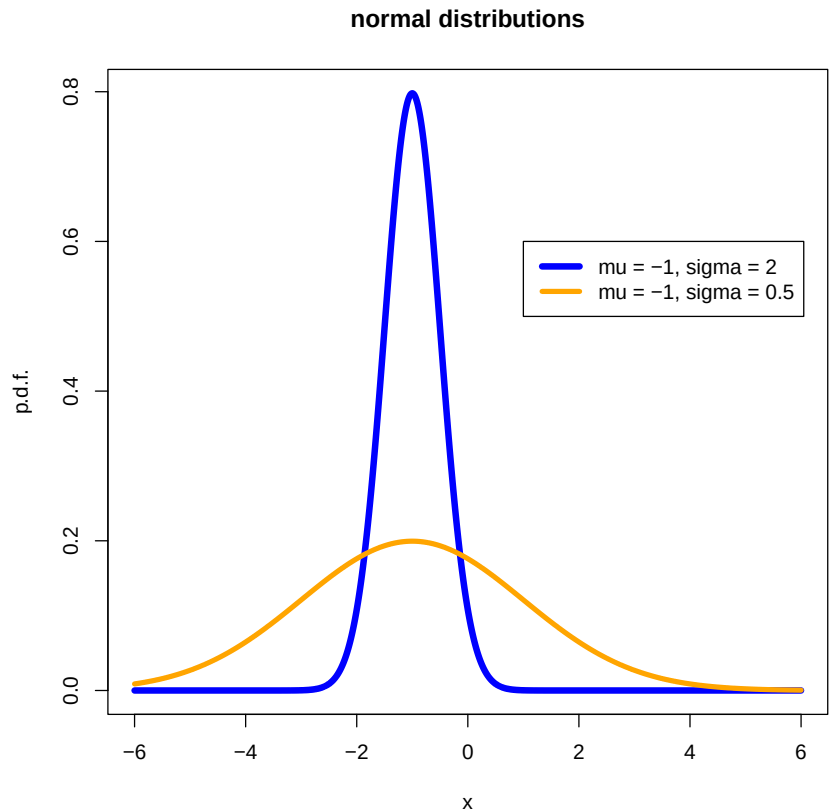
```
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE,
log.p = FALSE)
```

- Quantile function

```
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE,
log.p = FALSE)
```

Normal distributions





- Left: distributions with $\mu = 0, \sigma = 1$ and $\mu = 2, \sigma = 1$.
- Right: distributions with $\mu = -1, \sigma = 2$ and $\mu = -1, \sigma = \frac{1}{2}$.

Priors for the normal model

- The normal distribution is the benchmark model for continuous random variables that can be positive or negative.

$$y_i \sim \mathrm{N}\left(\mu, \sigma^2\right)$$

- If only the mean μ is unknown, the conjugate prior is the normal distribution

$$\mu \sim \mathrm{N}\left(\underline{\mu}, \underline{\kappa}\right)$$

Priors for the normal model

- The normal distribution is the benchmark model for continuous random variables that can be positive or negative.

$$y_i \sim \mathrm{N}\left(\mu, \sigma^2\right)$$

- If both the mean μ and variance σ^2 are unknown, we need a joint prior distribution

$$p\left(\boldsymbol{\mu}, \sigma^2\right) = p\left(\boldsymbol{\mu}|\sigma^2\right) p\left(\sigma^2\right)$$

- The conjugate prior is

$$\begin{aligned} \mu &\sim \mathrm{N}\left(\underline{\mu}, \underline{\kappa}\sigma^2\right) \\ \sigma^2 &\sim \text{Inv Gamma}\left(\underline{\alpha}, \underline{\delta}\right) \end{aligned}$$

which we will discuss later.

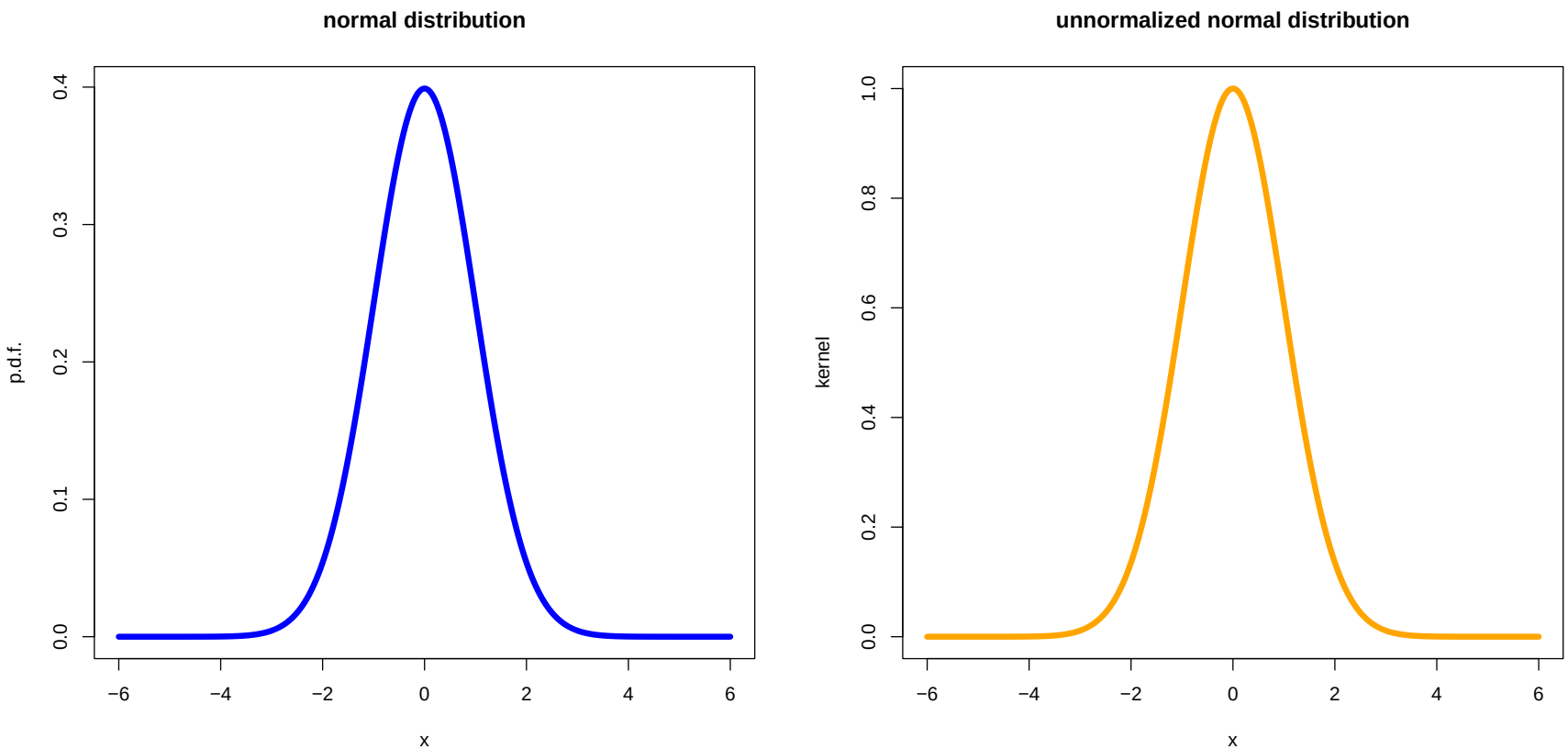
"Up to proportionality"

- In many situations, a probability distribution for a parameter μ is only known up to proportionality.

$$p(\mu|\mathbf{y}, \sigma^2) \propto p(\mathbf{y}|\mu, \sigma^2)p(\mu)$$

- The mathematical notation "$\propto$" means "proportional to".
- We can ignore the normalizing constant $C = p(\mathbf{y})$ and only look at the posterior kernel $k(\mu|\mathbf{y}, \sigma^2)$.
- The shape of the distribution $p(\mu|\mathbf{y}, \sigma^2)$ does not depend on C because it is not a function of μ (or σ^2).

Normalized vs. unnormalized distributions



- The p.d.f. $p(\mu)$ and the kernel function $k(\mu)$ have the exact same shape.
- The area under the curve on the left equals 1.

Prior distribution

- Our prior is

$$\mu \sim \mathsf{N}\left(\underline{\mu}, \underline{\kappa}\right)$$

- Notation: I use "underlines" like $\underline{\mu}$ to emphasize that they are prior **hyperparameters**.
  - $\underline{\mu}$ is the prior mean
  - $\underline{\kappa}$ is the prior variance
  - $\underline{\kappa}^{-1}$ is the prior precision
  - We select $\underline{\mu}$ and $\underline{\kappa}$ based on prior knowledge or beliefs.

Error precision

- The **precision** of a random variable is defined as the inverse of the variance.
- If σ^2 is the variance, then the precision is $h = \frac{1}{\sigma^2}$.
  - A small variance has a large precision.
- I write the model in terms of the variance

$$\mu \sim \mathsf{N}\left(\underline{\mu}, \underline{\kappa}\right)$$

- Stats and machine learning often use the precision

$$\mu \sim \mathsf{N}\left(\underline{\mu}, \underline{\kappa}^{-1}\right)$$

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Likelihood function

- The **likelihood function** is the joint probability distribution of the data when μ, σ^2 are known.
- Assuming $\{y_i\}_{i=1}^n$ are i.i.d., then

$$\begin{aligned} p\left(y_1, \ldots, y_n | \mu, \sigma^2\right) &= \prod_{i=1}^{n} p\left(y_i | \mu, \sigma^2\right) \\ &= \prod_{i=1}^{n} \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) \exp\left(-\frac{1}{2\sigma^2}\left[y_i - \mu\right]^2\right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[y_i - \mu\right]^2\right) \end{aligned}$$

- This is a function of μ (and σ^2 if it is also unknown).

Comments

- The distribution $p\left(y_i|\mu,\sigma^2\right)$ is a probability distribution for y_i when the parameters μ and σ^2 are known.
- But, we face the opposite situation.
- We observe the data $\mathbf{y} = (y_1, \ldots, y_N)^\top$.
- The data $\mathbf{y}$ is known.
- We do not know the parameters μ or σ^2 that generated the data.
- The likelihood function is not a probability distribution for μ or σ^2

Log-Likelihood function

- The likelihood function is

$$p\left(\mathbf{y}|\mu,\sigma^2\right) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[y_i-\mu\right]^2\right)$$

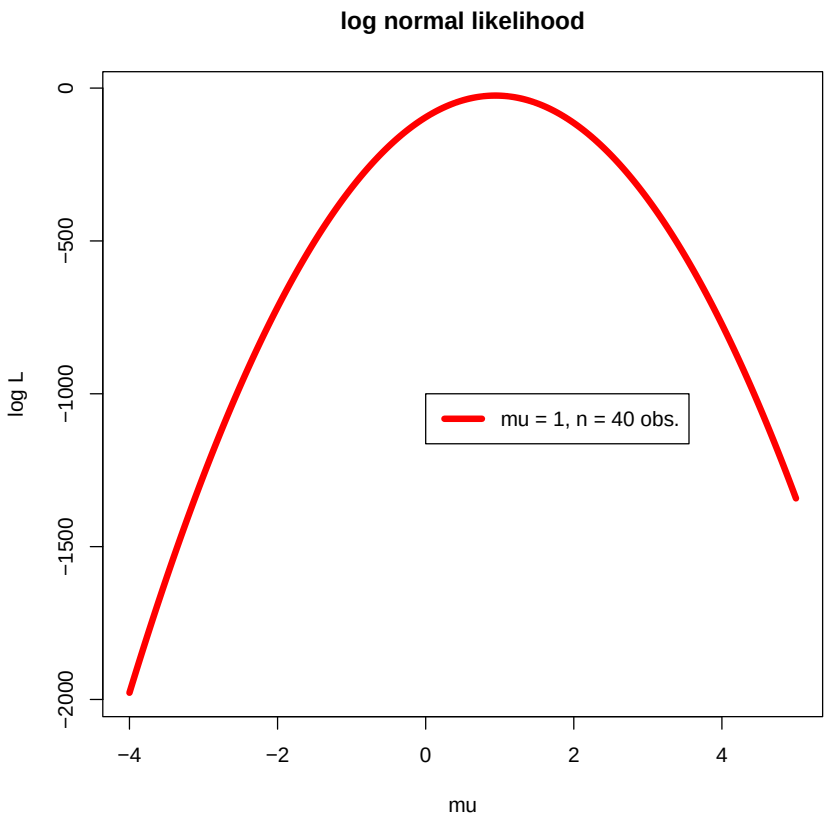
- Take (natural) logarithms

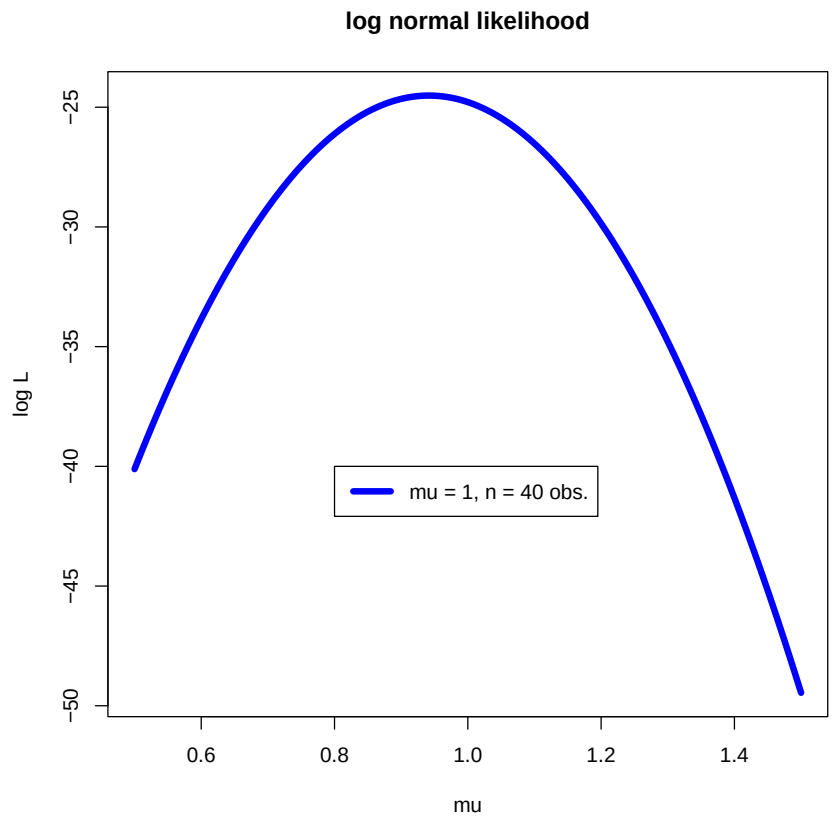
$$\log p\left(\mathbf{y}|\mu,\sigma^2\right) = -\frac{n}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y_i-\mu\right)^2$$

- This is a function of μ and σ^2.

- If you want, you can ignore the normalizing constants.

Log-likelihood function of μ





- $n = 50$ and true parameter is $\mu = 1$
- Plots show the log-likelihood function (quadratic).

R code

- Plotting the log-likelihood function

```
set.seed(12)
mu_true = 1
sigma_true = 0.5
n = 40
y = rnorm(n,mu_true,simga_true)
mu_seq = seq(-4,5,0.001)
iL = length(mu_seq)
# calculate the log-likelihood
logLikes = matrix(0,iL,1)
for(i in 1:iL){
    logLikes[i] =
sum(dnorm(y,mu_seq[i],sigma_true,log=TRUE))
 }
```

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Comments

- We want to calculate the **maximum likelihood estimator** for the normal distribution.
- This is a benchmark frequentist estimator.
- We do this by finding the values of μ and σ^2 that maximize the log-likelihood function.
- To do this, we take the derivatives of the log-likelihood function with respect to μ and σ^2.
  - We set the "first order conditions" equal to zero.
  - We solve 2 equations in 2 unknowns.

MLE for the normal distribution

- The log-likelihood function is

$$\log p\left(\mathbf{y}|\mu,\sigma^2\right) = -\frac{n}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y_i-\mu\right)^2$$

- The partial derivative w.r.t. to μ is

$$\frac{\partial \log p\left(\mathbf{y}|\mu,\sigma^2\right)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}\left(y_i-\mu\right)$$

- The partial derivative w.r.t. to σ^2 is

$$\frac{\partial \log p\left(\mathbf{y}|\mu,\sigma^2\right)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}\left(y_i-\mu\right)^2$$

MLE for the normal distribution

- Set the first F.O.C. for μ equal to zero

$$
\begin{aligned}
0 &= \frac{1}{\sigma^2} \sum_{i=1}^{n} (y_i - \mu) \\
0 &= \sum_{i=1}^{n} (y_i - \mu) \\
0 &= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \mu \\
0 &= \sum_{i=1}^{n} y_i - n\mu \\
\mu &= \frac{1}{n} \sum_{i=1}^{n} y_i
\end{aligned}
$$

- MLE of μ is the sample mean $\widehat{\mu} = \frac{1}{n} \sum_{i=1}^{n} y_i$ or $\bar{y}$.

MLE for the normal distribution

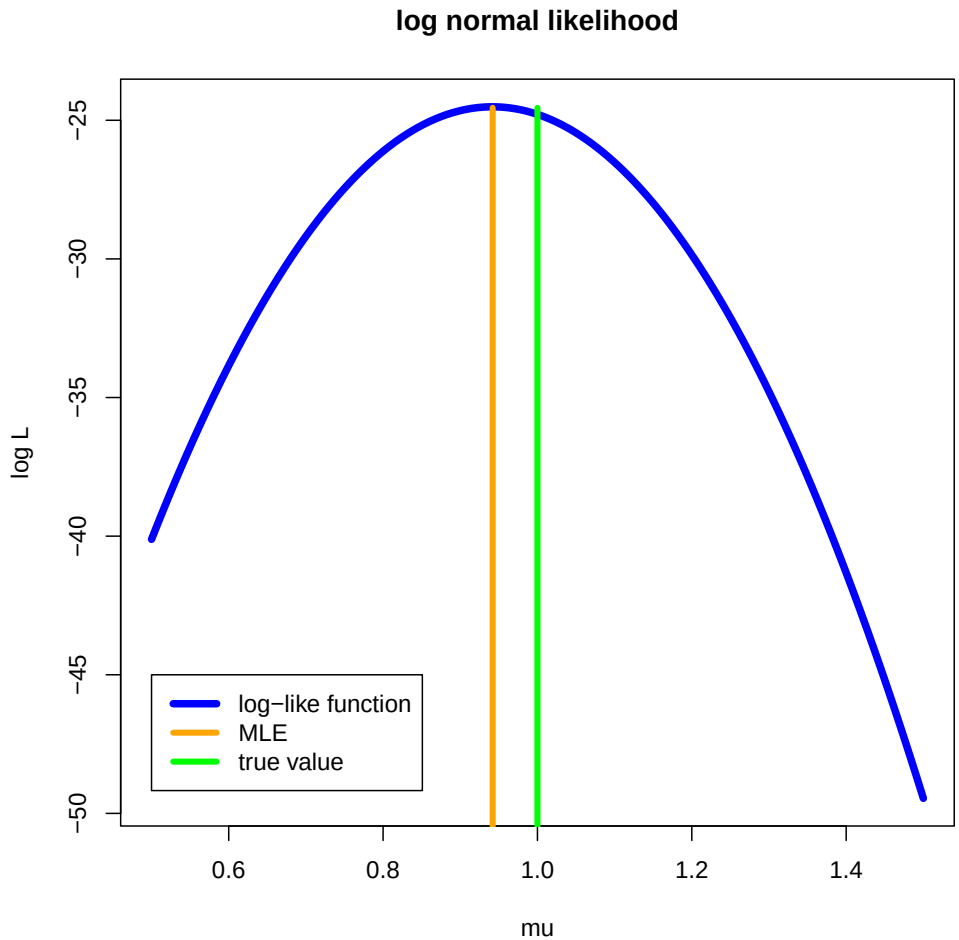
- Set the second F.O.C. for σ^2 equal to zero

$$\begin{aligned} 0 &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i-\mu)^2 \\ \frac{n}{2\sigma^2} &= \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i-\mu)^2 \\ \frac{\sigma^4}{\sigma^2} &= \frac{1}{n}\sum_{i=1}^{n}(y_i-\mu)^2 \\ \sigma^2 &= \frac{1}{n}\sum_{i=1}^{n}(y_i-\mu)^2 \end{aligned}$$

- MLE is the sample variance:

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i-\widehat{\mu})^2$$

Log-Likelihood function



- the maximum value is $\widehat{\mu}_{mle} = 0.91$
- True value is $\mu = 1$ and σ^2 is known.

Comments

- The maximum likelihood estimator is a classic approach to estimating an unknown parameter.
- Since it is based on the likelihood function, it is usually close to the Bayesian posterior mean or mode.
- In some complicated models, it can be easier to calculate the Bayesian posterior mean instead of the M.L.E.

Frequentist standard errors

- Remember that a frequentist's view of uncertainty is in "hypothetical repeated samples" of the data.
  - What if we observed a different sample?

- The variability of the MLE is

$$\begin{aligned}
\mathbb{V}\left(\widehat{\mu}_{mle}\right) &= \mathbb{V}\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right) = \frac{1}{n^2}\mathbb{V}\left(\sum_{i=1}^{n} y_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{V}\left(y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 \\
&= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}
\end{aligned}$$

- The frequentist standard error for the mean is $S.E.\left(\widehat{\mu}_{mle}\right) = \frac{\sigma}{\sqrt{n}}$

Feasible standard errors

- The frequentist standard error for the mean is

$$S.E.\left(\widehat{\mu}_{mle}\right) = \frac{\sigma}{\sqrt{n}}$$

- The answer depends on a parameter σ that we do not know.

- The solution is to approximate the unknown σ using the sample standard deviation

$$S.E.\left(\widehat{\mu}_{mle}\right) = \frac{\widehat{\sigma}}{\sqrt{n}}$$

given by

$$\widehat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{\mu}\right)^2}$$

Frequentist standard errors

- One can also show that the variability of the MLE for σ^2 is

$$\mathbb{V}\left(\widehat{\sigma}^2_{mle}\right) = \frac{\sigma^4}{n}$$

- Again, this depends on the unknown parameter.
- We can approximate it using the estimator

$$S.E.\left(\widehat{\sigma}^2_{mle}\right) = \sqrt{\frac{\widehat{\sigma}^4}{n}} = \frac{\widehat{\sigma}^2}{\sqrt{n}}$$

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Posterior distribution

- The posterior distribution $p\left(\mu|\mathbf{y},\sigma^2\right)$ is

$$\begin{aligned} p\left(\mu|\mathbf{y},\sigma^2\right) &= \frac{p\left(\mathbf{y}|\mu,\sigma^2\right)p\left(\mu\right)}{p\left(\mathbf{y}\right)} \\ &= \frac{\text{likelihood} \times \text{prior}}{\text{normalizing constant}} \end{aligned}$$

- We can ignore the normalizing constant

$$\begin{aligned} p\left(\mu|\mathbf{y},\sigma^2\right) &\propto p\left(\mathbf{y}|\mu,\sigma^2\right)p\left(\mu\right) \\ &\propto \text{likelihood} \times \text{prior} \end{aligned}$$

- $\mathbf{y} = \left(y_1,\ldots,y_N\right)^\top$

Posterior distribution

- The posterior distribution $p\left(\mu|\mathbf{y},\sigma^2\right)$ is

$$p\left(\mu|\mathbf{y},\sigma^2\right) \propto \underbrace{\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[y_i-\mu\right]^2\right)}_{\text{likelihood}}\underbrace{\exp\left(-\frac{1}{2\underline{\kappa}}\left[\mu-\underline{\mu}\right]^2\right)}_{\text{prior for }\mu}$$

- We will show that this is the posterior kernel of the normal distribution.
- To do this, we need to "complete the square."
- I will derive this in class (you do not need to know this for an exam).

Posterior distribution

- This is the posterior kernel of a normal distribution with updated parameters.

$$p\left(\mu|\mathbf{y},\sigma^2\right) \quad \propto \quad \exp\left(-\frac{1}{2\overline{\kappa}}\left(\mu-\overline{\mu}\right)^2\right)$$

- We find

$$p\left(\mu|\mathbf{y},\sigma^2\right) \quad = \quad \mathsf{N}\left(\overline{\mu},\overline{\kappa}\right)$$

where

$$\begin{aligned}\overline{\kappa}^{-1} &= \underline{\kappa}^{-1}+\frac{n}{\sigma^2}\\ \overline{\mu} &= \overline{\kappa}\left(\underline{\kappa}^{-1}\underline{\mu}+\frac{1}{\sigma^2}\sum_{i=1}^{n}y_i\right)\end{aligned}$$

Posterior mean

- The posterior is

$$p\left(\mu|\mathbf{y}\right) \quad = \quad \mathsf{N}\left(\overline{\mu}, \overline{\kappa}\right)$$

with posterior mean

$$\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) \quad = \quad \overline{\kappa}\left(\underline{\kappa}^{-1}\underline{\mu} + \frac{1}{\sigma^2}\sum_{i=1}^{n} y_i\right)$$

- We can rearrange the posterior mean as

$$\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) \quad = \quad \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1} + \frac{n}{\sigma^2}} \underbrace{\underline{\mu}}_{\text{prior mean}} + \frac{\frac{n}{\sigma^2}}{\underline{\kappa}^{-1} + \frac{n}{\sigma^2}} \times \underbrace{\frac{1}{n}\sum_{i=1}^{n} y_i}_{\text{M.L.E.}}$$

- The posterior mean is a weighted average of the prior mean and maximum likelihood estimator.

Posterior mean

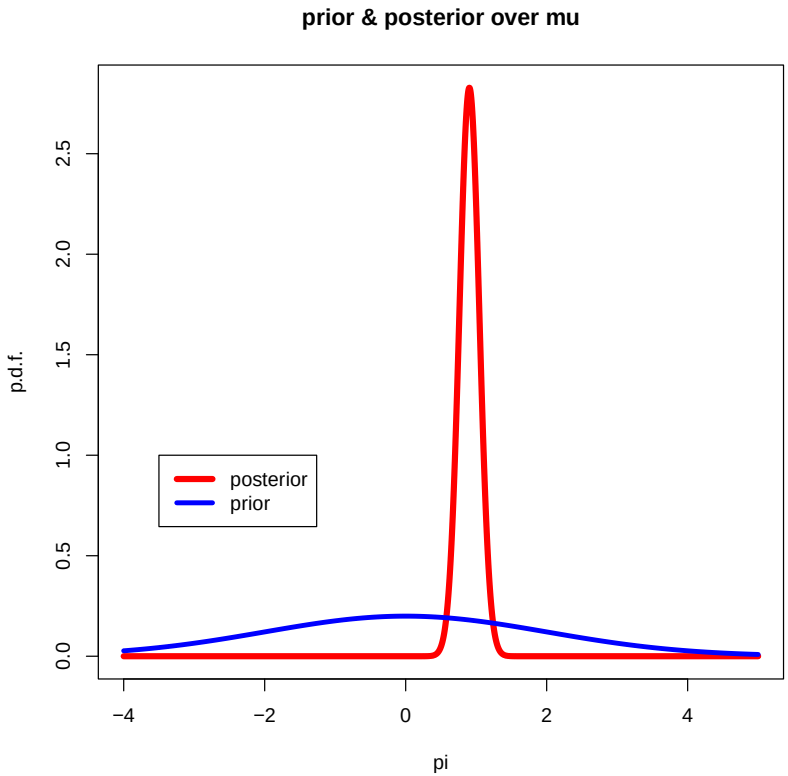
- The posterior mean is

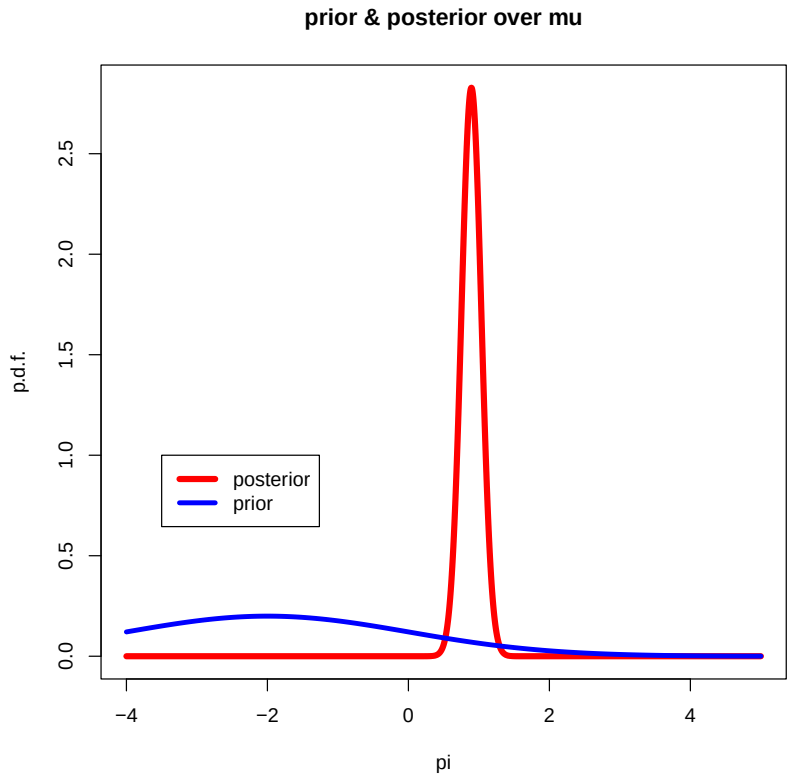
$$\mathbb{E}\left(\mu|\mathbf{y},\sigma^2\right) = \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1}+\frac{n}{\sigma^2}} \underbrace{\underline{\mu}}_{\text{prior mean}} + \frac{\frac{n}{\sigma^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma^2}} \times \underbrace{\frac{1}{n}\sum_{i=1}^{n} y_i}_{\text{M.L.E.}}$$

- What happens when....
  - the sample size n gets large, i.e. $n \longrightarrow \infty$?
  - the prior precision $\underline{\kappa}^{-1}$ goes to zero? Or, equivalently, the prior variance $\underline{\kappa}$ goes to infinity?

- Normal distribution: posterior mean = posterior mode

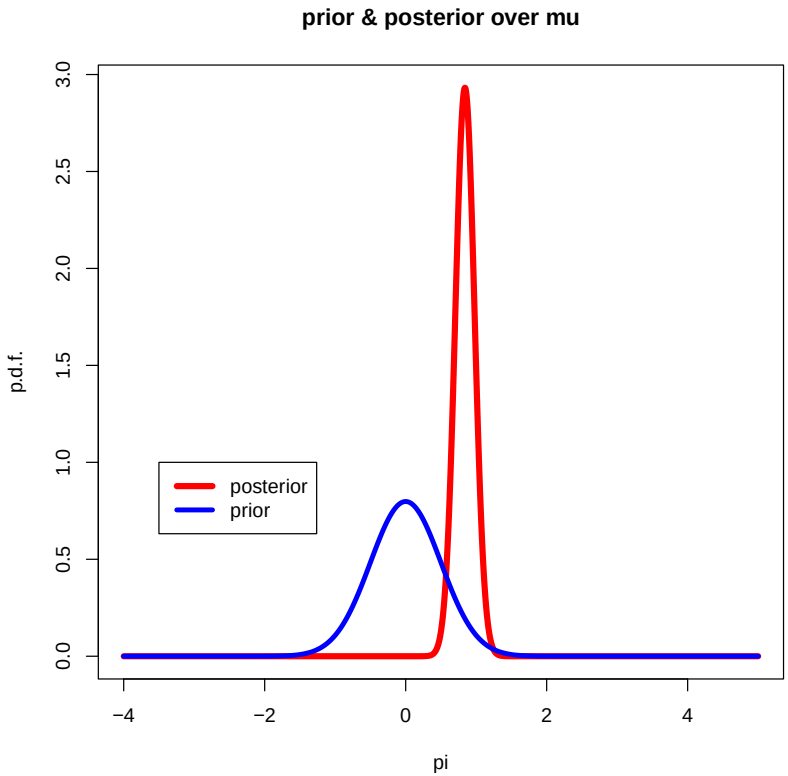
Posterior distribution

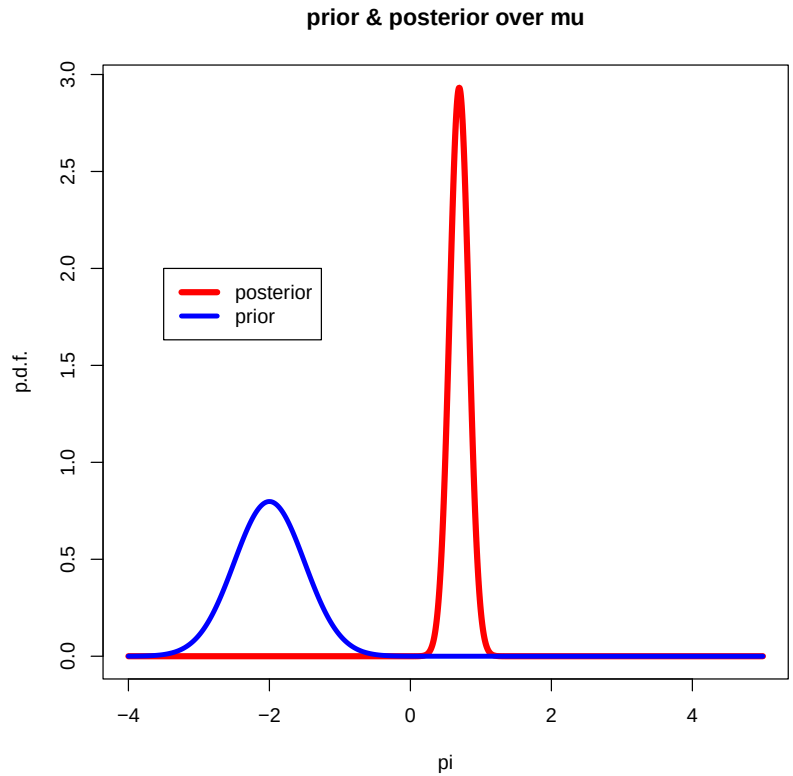




- $n = 50$, true parameter $\mu = 0.7$, $\widehat{\mu}_{mle} = 0.908$
- Left: prior with $\underline{\mu} = 0$ and $\underline{\kappa} = 4$, $\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) = 0.904$
- Right: prior with $\underline{\mu} = -2$ and $\underline{\kappa} = 4$, $\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) = 0.893$

Posterior distribution





- $n = 50$, true parameter $\mu = 0.7$, $\widehat{\mu}_{mle} = 0.908$
- Left: prior with $\underline{\mu} = 0$ and $\underline{\kappa} = 0.25$, $\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) = 0.841$
- Right: prior with $\underline{\mu} = -2$ and $\underline{\kappa} = 0.25$, $\mathbb{E}\left(\mu|\mathbf{y}, \sigma^2\right) = 0.693$

Posterior variance and standard deviation

- The posterior is

$$p\left(\mu|\mathbf{y},\sigma^2\right) \quad = \quad \mathsf{N}\left(\overline{\mu},\overline{\kappa}\right)$$

- The posterior variance is

$$\mathbb{V}\left(\mu|\mathbf{y},\sigma^2\right) \quad = \quad \overline{\kappa} \quad = \quad \frac{1}{\underline{\kappa}^{-1}+\frac{n}{\sigma^2}} \quad = \quad \frac{1}{\frac{\sigma^2}{n}\underline{\kappa}^{-1}+1} \times \left(\frac{\sigma^2}{n}\right)$$

- We typically report the **posterior standard deviation** as a measure of uncertainty
  - This is analogous to the frequentist's standard error.
  - We derived the frequentist standard error: $\sqrt{\frac{\sigma^2}{n}}$.
  - How do these compare?

R code

- We can calculate

```
mu_true = 0.7
sigma2_true = 1

set.seed(12)
n = 50;
y = rnorm(n,mu_true,sqrt(sigma2_true))
y_bar = mean(y)

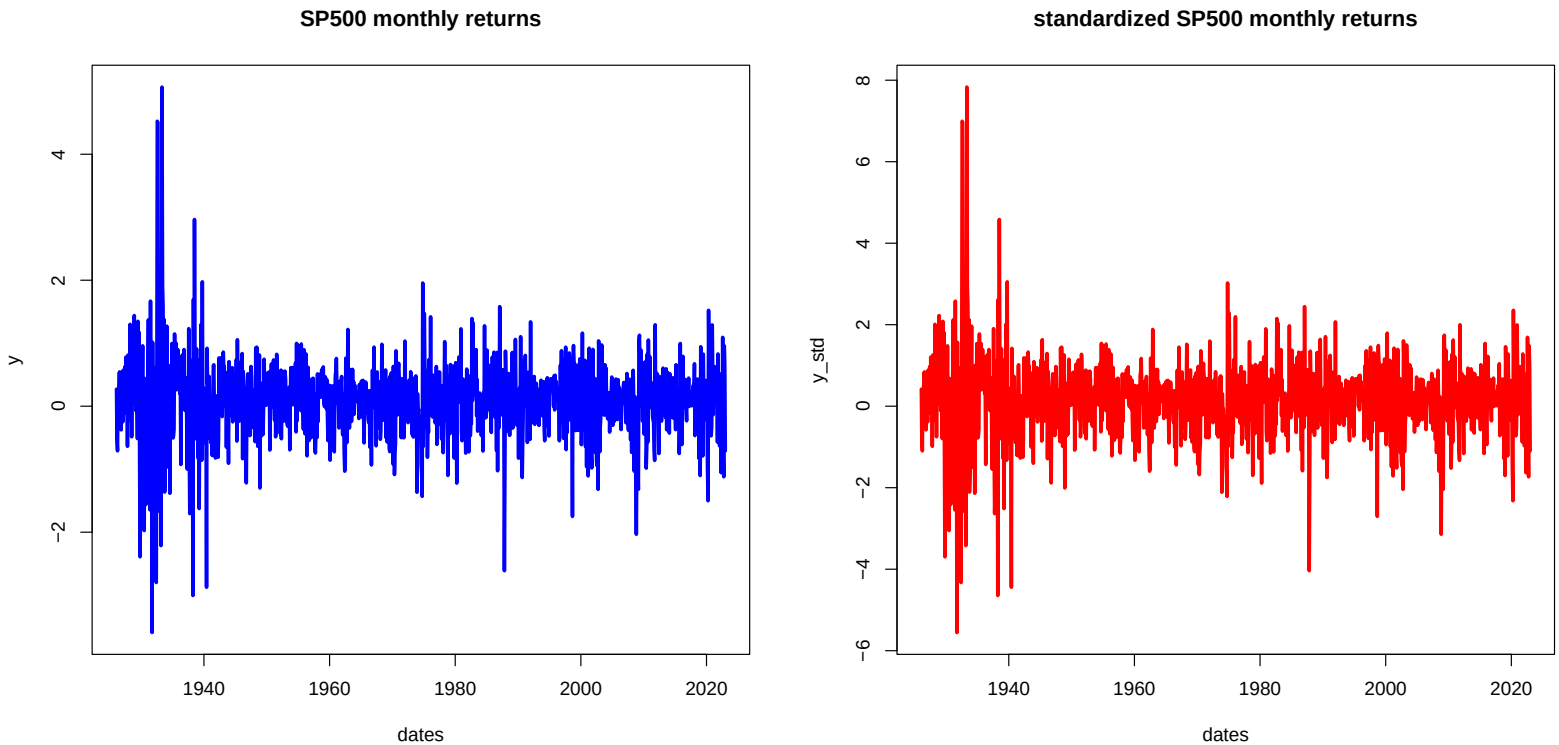
mu_prior = 0
kappa_prior = 4

kappa_post = 1/(1/kappa_prior + n/sigma2_true)
mu_post = kappa_post*( mu_prior/kappa_prior +
(n/sigma2)*y_bar)

iNumSim = 100000
mu_sim = rnorm(iNumSim,mu_post,sqrt(kappa_post))

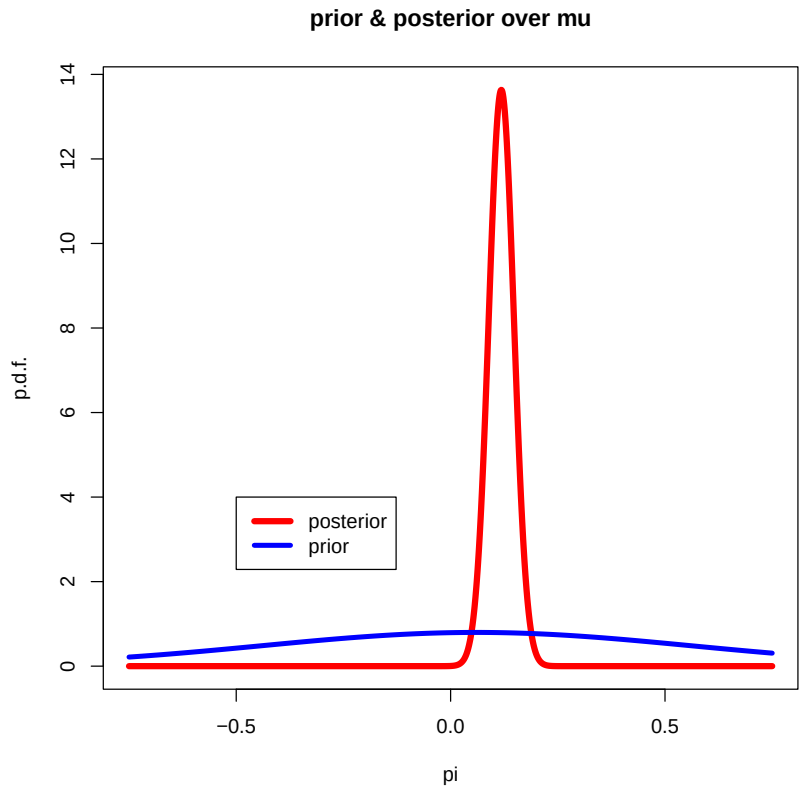
mean(mu_sim)
var(mu_sim)
sd(mu_sim)
```

Ex: S&P500 monthly returns



- $n = 1164$, 1/1926 to 12/2022
- $\widehat{\mu}_{mle} = 0.0766$ and $\widehat{\sigma} = 0.646$
- Left: monthly returns (annualized); Right: standardized

Ex: S&P500 monthly returns



- Using the standardized data (because σ^2 is known)
- $\widehat{\mu}_{mle} = 0.1181$ and $S.E.\left(\widehat{\mu}_{mle}\right) = 0.0293$
- Posterior: $\mathbb{E}\left(\mu|\mathbf{y}\right) = 0.1180$ and $S.D.\left(\mu|\mathbf{y}\right) = 0.0292$

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Bias-variance trade off

- Recall that the mean square error can be decomposed as the variance of the estimator plus the squared bias.

$$m.s.e.\left(\widehat{\theta}\right) \quad = \quad \mathbb{V}\left(\widehat{\theta}\right) + \left(\textbf{bias}\left(\widehat{\theta}\right)\right)^2$$

- Let's compare the MSE of the Bayesian estimator versus the MLE.

- Since the MLE is the sample mean, we know that it is unbiased.

MSE of the MLE

- The MLE has bias and variance

$$\begin{aligned} \mathbb{E}\left(\widehat{\mu}_{m.l.e.}\right) - \mu_0 &= 0 \\ \mathbb{V}\left(\widehat{\mu}_{m.l.e.}\right) &= \frac{\sigma_0^2}{n} \end{aligned}$$

  where μ_0 and σ_0^2 are the "true" parameters.

- The MSE is therefore

$$m.s.e.\left(\widehat{\mu}_{m.l.e.}\right) = \frac{\sigma_0^2}{n}$$

- Note: this is not a function of the true μ_0.

Bias of the Bayesian posterior mean

- The bias is

$$\begin{aligned}
\mathbb{E}\left(\widehat{\mu}_{bayes}\right) - \mu_0 &= \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}}\underline{\mu} + \frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}}\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right) - \mu_0 \\
&= \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}}\underline{\mu} + \frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}}\mu_0 - \mu_0 \\
&= \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}}\underline{\mu} + \left(\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1} + \frac{n}{\sigma_0^2}} - 1\right)\mu_0
\end{aligned}$$

- There is no bias if $\underline{\kappa}^{-1} = 0$, i.e. a diffuse prior.

- The bias goes to zero as $n \to \infty$.

Variance of the Bayesian posterior mean

- The (frequentist) variance of the posterior mean is

$$\begin{aligned}
\mathbb{V}\left(\widehat{\mu}_{bayes}\right) &= \mathbb{V}\left(\frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\underline{\mu}+\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\frac{1}{n}\sum_{i=1}^{n}y_i\right) \\
&= \left(\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\right)^2 \mathbb{V}\left(\frac{1}{n}\sum_{i=1}^{n}y_i\right) \\
&= \left(\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\right)^2 \frac{\sigma_0^2}{n}
\end{aligned}$$

- Recall that the variance of the MLE is $\frac{\sigma_0^2}{n}$.
- If $\underline{\kappa}^{-1} > 0$, variance is smaller than the variance of the MLE.

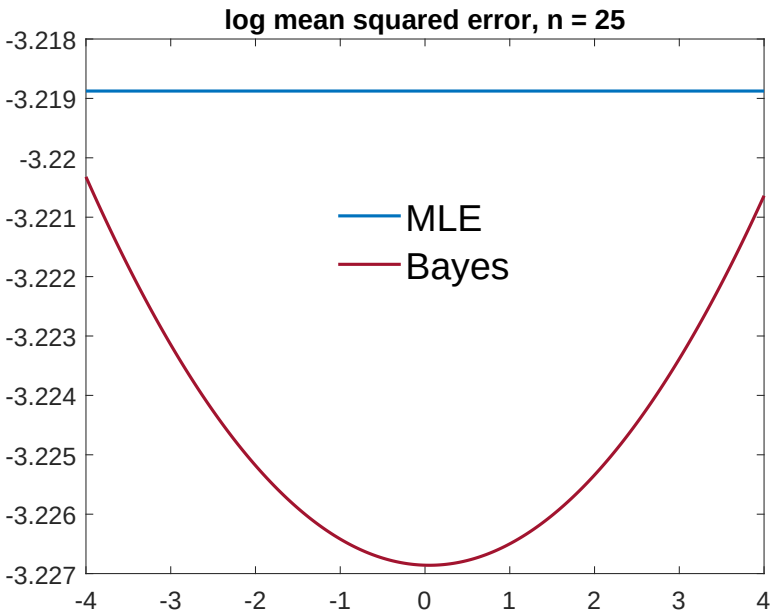
MSE of the Bayesian posterior mean

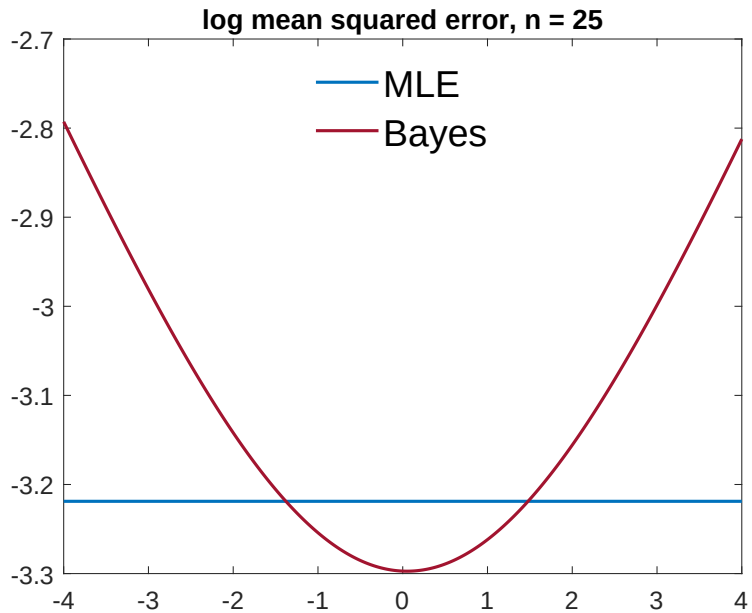
- The MSE of the Bayes estimator is

$$m.s.e.\left(\widehat{\mu}_{bayes}\right) = \left[\frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\underline{\mu}+\left(\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}-1\right)\mu_0\right]^2 + \left(\frac{\frac{n}{\sigma_0^2}}{\underline{\kappa}^{-1}+\frac{n}{\sigma_0^2}}\right)^2\frac{\sigma_0^2}{n}$$

- Again, this uses the frequentist notion of repeated sampling.

log-MSE as a function of μ_0





- Left: $n = 25$, $\underline{\mu} = 0.05$, $\underline{\kappa} = 10$
- Right: $n = 25$, $\underline{\mu} = 0.05$, $\underline{\kappa} = 1$.
- Bayes estimator has lower MSE for a wide range of values of μ_0 but can be sensitive to the prior variance $\underline{\kappa}$.

Bias variance trade-off

- Adding prior information adds bias but reduces the variance.
  - Be careful. Do not choose a small prior variance unless you are certain that the mean is a narrow range.
- In high dimensions, reducing the variance becomes more and more important.

Outline

Prior: normal distribution

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Posterior distribution: unknown mean and known variance

Bias variance trade-off

Prior: normal inverse gamma distribution

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Highest posterior density intervals

Marginal likelihood

Posterior predictive distribution

Gamma distribution

Definition

A random variable x has a **gamma distribution** written $x \sim \text{Gamma}(\alpha, \delta)$, if the probability density function (p.d.f.) is

$$p(x|\alpha, \delta) = \frac{1}{\Gamma(\alpha)\delta^{\alpha}} x^{\alpha-1} \exp\left(-\frac{x}{\delta}\right)$$

where α is the shape parameter and δ is the scale parameter.

- $\mathbb{E}[x] = \alpha\delta$.
- $\mathbb{V}[x] = \alpha\delta^2$.
- Kernel: $k(x) = x^{\alpha-1} \exp\left(-\frac{x}{\delta}\right)$.
- Normalizing constant: $\frac{1}{\Gamma(\alpha)\delta^{\alpha}}$

Expected value

- We can derive the expected value of the distribution

$$
\begin{aligned}
\mathbb{E}(x) &= \int_0^\infty x p(x)\, dx \\
&= \int_0^\infty x \frac{1}{\Gamma(\alpha)\,\delta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\delta}\right) dx \\
&= \frac{1}{\Gamma(\alpha)\,\delta^\alpha} \int_0^\infty \underbrace{x^{\alpha+1-1} \exp\left(-\frac{x}{\delta}\right)}_{\textit{kernel of gamma}} dx \\
&= \frac{1}{\Gamma(\alpha)\,\delta^\alpha} \Gamma(\alpha+1)\,\delta^{\alpha+1} \\
&= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \delta \\
&= \frac{\Gamma(\alpha)}{\Gamma(\alpha)} \alpha\delta \;=\; \alpha\delta
\end{aligned}
$$

Mode

- We can derive the mode of the distribution

$$\log p\left(x|\alpha,\delta\right) = -\log\left[\Gamma\left(\alpha\right)\right] - \alpha\log\left(\delta\right) + \left(\alpha-1\right)\log\left(x\right) - \frac{x}{\delta}$$

- Take the derivative with respect to x

$$\frac{d\log p\left(x|\alpha,\delta\right)}{dx} = \frac{\left(\alpha-1\right)}{x} - \delta$$

- Set this F.O.C. equal to zero and solve

$$\begin{aligned}
\frac{\left(\alpha-1\right)}{x} - \frac{1}{\delta} &= 0 \\
\frac{\left(\alpha-1\right)}{x} &= \frac{1}{\delta} \\
x &= \left(\alpha-1\right)\delta
\end{aligned}$$

R code

- Probability density function

```
dgamma(x, shape, rate = 1, scale = 1/rate, log =
FALSE)
```

- Generate random numbers

```
rgamma(n, shape, rate = 1, scale = 1/rate)
```

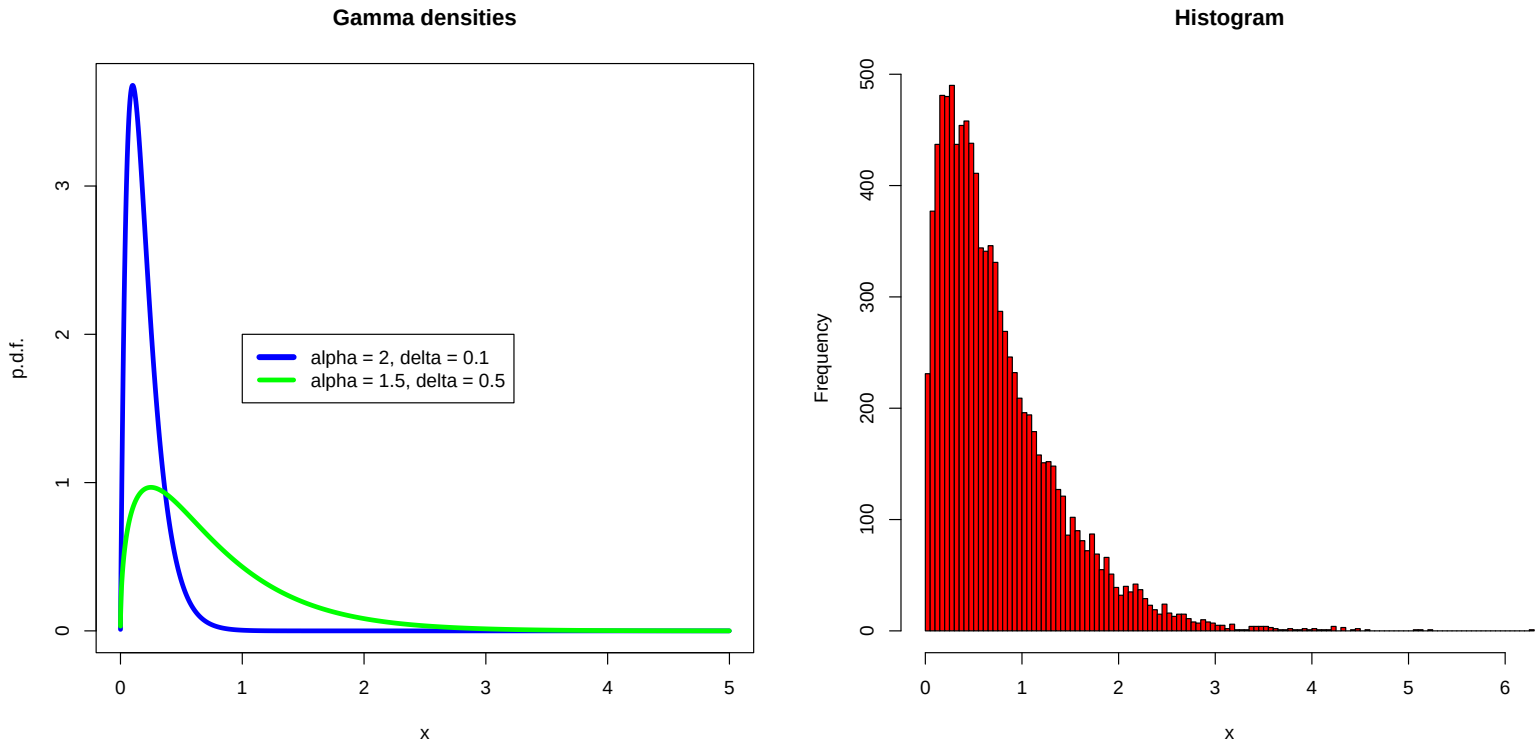
- Cumulative distribution function

```
pgamma(q, shape, rate = 1, scale = 1/rate,
lower.tail = TRUE, log.p = FALSE)
```

- Quantile function

```
qgamma(p, shape, rate = 1, scale = 1/rate,
lower.tail = TRUE, log.p = FALSE)
```

Gamma densities



- Left: probability density functions
- Right: histogram of simulated values
- Note: a gamma random variable must be positive!

Comments

- A gamma distribution is common for random variables that must be positive.
- The Chi-square χ^2_k distribution is a special case of the gamma distribution.
  - If $x \sim \chi^2_k$, then

$$x \sim \text{Gamma}\left(\frac{k}{2}, 2\right)$$

- The exponential distribution is a special case of the gamma distribution
  - If $x \sim \text{Exponential}(\lambda)$, then

$$x \sim \text{Gamma}(1, \lambda)$$

Comments

- Some authors and researchers define the gamma distribution in terms of the "**rate**" instead of the "**scale**" parameter.
- For the gamma distribution:

$$\begin{aligned} \text{rate} &= \frac{1}{\text{scale}} \\ \text{scale} &= \frac{1}{\text{rate}} \end{aligned}$$

- In statistics, it is more common to use the rate.
- In econometrics, it is more common to use the scale.
- Unfortunate source of confusion.

Comments

- If you define the distribution in terms of the scale parameter δ, the p.d.f. is

$$p\left(x|\alpha,\delta\right) = \frac{1}{\Gamma\left(\alpha\right)\delta^{\alpha}}x^{\alpha-1}\exp\left(-\frac{x}{\delta}\right)$$

with mean $\mathbb{E}\left(x\right) = \alpha\delta$.

- If you define the distribution in terms of the rate parameter $\gamma = \frac{1}{\delta}$, the p.d.f. is

$$p\left(x|\alpha,\gamma\right) = \frac{\gamma^{\alpha}}{\Gamma\left(\alpha\right)}x^{\alpha-1}\exp\left(-\gamma x\right)$$

with mean $\mathbb{E}\left(x\right) = \frac{\alpha}{\gamma}$.

Inverse gamma distribution

Definition

A random variable x has an **inverse gamma distribution** written $x \sim \text{Inv Gamma}(\alpha, \delta)$, if the probability density function (p.d.f.) is

$$p(x|\alpha, \delta) = \frac{\delta^{\alpha}}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\delta}{x}\right)$$

where α is the shape parameter and δ is the scale parameter.

- $\mathbb{E}[x] = \frac{\delta}{\alpha-1}$.
- Kernel: $k(x) = x^{-\alpha-1} \exp\left(-\frac{\delta}{x}\right)$.
- Normalizing constant: $\frac{\delta^{\alpha}}{\Gamma(\alpha)}$

Comments

- The name "inverted" gamma is literal.
- If $X \sim \text{Gamma}\left(\alpha, \delta\right)$, then $Y = \frac{1}{X}$ has distribution $Y \sim \text{Inv-Gamma}\left(\alpha, \frac{1}{\delta}\right)$.
- The converse is also true.
- If $Y \sim \text{Inv-Gamma}\left(\alpha, \delta\right)$, then $X = \frac{1}{Y}$ has distribution $X \sim \text{Gamma}\left(\alpha, \frac{1}{\delta}\right)$.
- The parameter α is the **shape**.
- The parameter δ is the **scale**.

Expected value

- If $\alpha > 1$, we can derive the expected value of the distribution

$$
\begin{aligned}
\mathbb{E}(x) &= \int_0^\infty x p(x)\, dx \\
&= \int_0^\infty x \frac{\delta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\delta}{x}\right) dx \\
&= \frac{\delta^\alpha}{\Gamma(\alpha)} \int_0^\infty \underbrace{x^{-(\alpha-1)-1} \exp\left(-\frac{\delta}{x}\right)}_{\text{kernel of inv. gamma}} dx \\
&= \frac{\delta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha-1)}{\delta^{\alpha-1}} \\
&= \frac{\Gamma(\alpha-1)}{\Gamma(\alpha)} \delta \\
&= \frac{\Gamma(\alpha-1)}{\Gamma(\alpha-1)} \frac{\delta}{\alpha-1} = \frac{\delta}{\alpha-1}
\end{aligned}
$$

R code

- Unfortunately, there are no built in `R` functions for the inverse gamma distribution.
- Unfortunately, the downloadable `R` package "`invgamma`" is not implemented correctly.
  - `library(invgamma)` is incorrect!
  - It defines the "scale" and "rate" parameters backwards.

R code

- To draw an inverse gamma random variable, we use the fact that if $x \sim$ Inv-Gamma (α, δ), then $y = \frac{1}{x}$ is

$$y \sim \text{Gamma}\left(\alpha, \frac{1}{\delta}\right)$$

where $\frac{1}{\delta}$ is the **scale** parameter of the Gamma distribution.

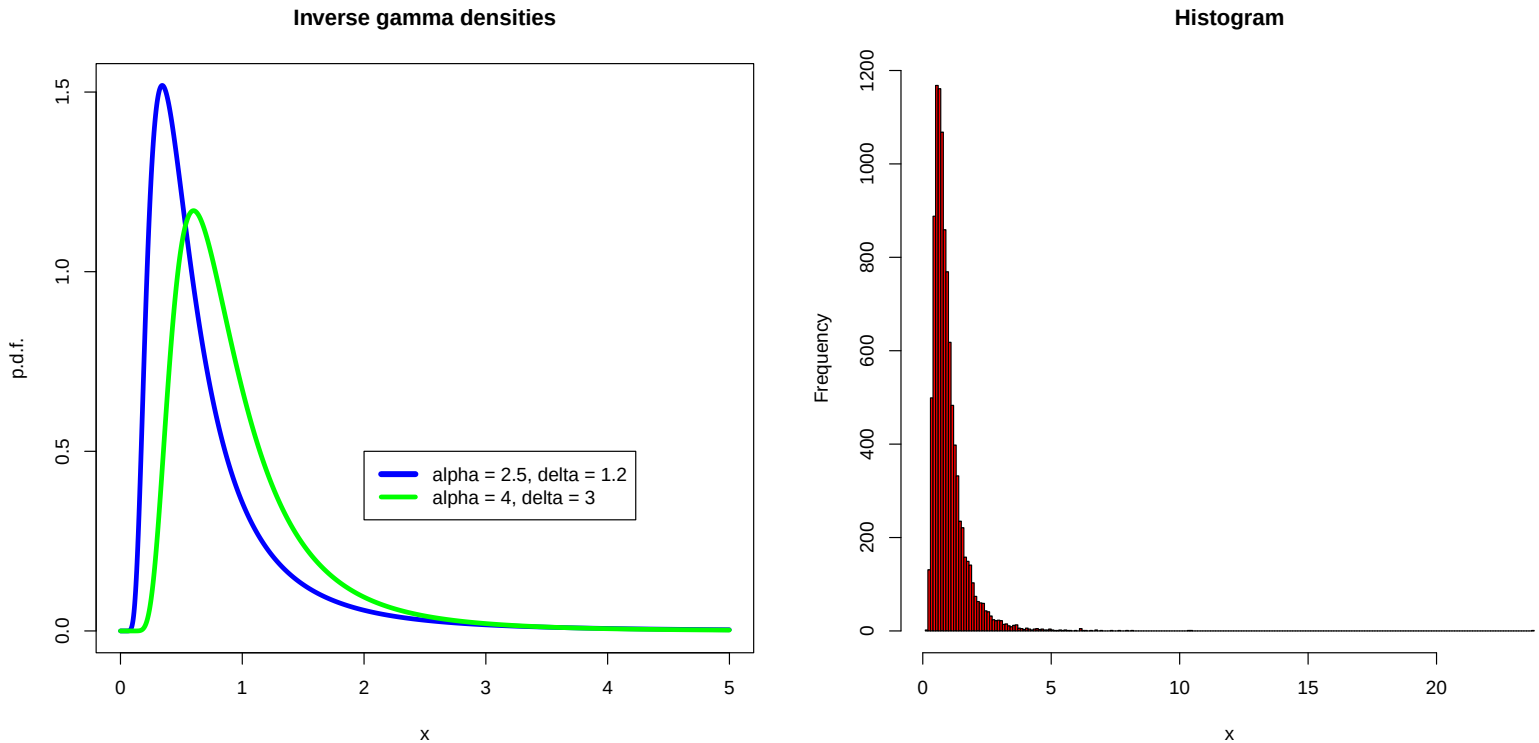
- Implementation:

```
x = 1/rgamma(1, shape = alpha, scale = 1/delta)
```

or

```
y = rgamma(1, shape = alpha, scale = 1/delta)
x = 1/y
```

Inverse gamma densities



- Left: probability density functions
- Right: histogram of simulated values

R code

- Plotting the probability density function

```
# parameters of the distribution
alpha = 4
delta = 2

# generate sequence of x values
x = seq(0.001,5,by=0.001)

# evaluate the probability density function
log_y = alpha*log(delta)-lgamma(alpha)
-(alpha+1)*log(x) - delta/x
y = exp(log_y)

# plot the p.d.f.
plot(x,y,type="l", col="red",lwd=5,main="Inverse
Gamma",ylab="p.d.f.")
```

Conjugate priors for the normal model

- If both the mean μ and variance σ^2 are unknown, the conjugate prior is

$$\begin{aligned} \mu &\sim \mathrm{N}\left(\underline{\mu}, \underline{\kappa}\sigma^2\right) \\ \sigma^2 &\sim \text{Inv Gamma}\left(\underline{\alpha}, \underline{\delta}\right) \end{aligned}$$

- The variance of the mean μ depends on the variance σ^2.

- The joint distribution is

$$p\left(\mu, \sigma^2\right) = p\left(\mu|\sigma^2\right) p\left(\sigma^2\right)$$

R: simulate from the prior

```
# prior for the variance
alpha_prior = 2.5
delta_prior = 1.2

# prior for the mean
mu_prior = 0.5
kappa_prior = 0.25

iNumDraws = 20000

# draw from the NIG distribution
sigma2_draws = 1/rgamma(iNumDraws, shape=alpha_prior, scale =
(1/delta_prior))

mu_draws = rnorm(iNumDraws, mu_prior,
sqrt(kappa_prior*sigma2_draws))

hist(mu_draws,breaks=200,freq=T,col="red",main=
paste("Histogram"),xlab="x")
```

Comments

- In statistics and machine learning, some textbooks/authors write the conjugate prior in terms of the precision $h = \frac{1}{\sigma^2}$.
- The prior is then

$$\begin{aligned} \mu &\sim \mathrm{N}\left(\underline{\mu}, \underline{\kappa}^{-1} h^{-1}\right) \\ h &\sim \text{Gamma}\left(\underline{\alpha}, \frac{1}{\underline{\delta}}\right) \end{aligned}$$

  where $\frac{1}{\underline{\delta}}$ is the **scale** parameter of the distribution.

- It is the same prior but uses different notation.
- This is very common in the literature.

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Posterior distribution

- The posterior distribution $p\left(\mu, \sigma^2 | \mathbf{y}\right)$ is

$$\begin{aligned} p\left(\mu, \sigma^2 | \mathbf{y}\right) &= \frac{p\left(\mathbf{y} | \mu, \sigma^2\right) p\left(\mu | \sigma^2\right) p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)} \\ &= \frac{\text{likelihood} \times \text{prior}}{\text{normalizing constant}} \end{aligned}$$

- We can ignore the normalizing constant

$$\begin{aligned} p\left(\mu, \sigma^2 | \mathbf{y}\right) &\propto p\left(\mathbf{y} | \mu, \sigma^2\right) p\left(\mu | \sigma^2\right) p\left(\sigma^2\right) \\ &\propto \text{likelihood} \times \text{prior} \end{aligned}$$

- $\mathbf{y} = \left(y_1, \ldots, y_N\right)^\top$

Posterior distribution

- The posterior distribution $p\left(\mu, \sigma^2 | \mathbf{y}\right)$ is

$$p\left(\mu, \sigma^2 | \mathbf{y}\right) \propto \underbrace{\left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left[y_i - \mu\right]^2\right)}_{\text{likelihood}}$$
$$\underbrace{\left(\frac{1}{\sigma^2}\right) \exp\left(-\frac{1}{2\underline{\kappa}\sigma^2}\left[\mu - \underline{\mu}\right]^2\right)}_{\text{prior for } \mu} \underbrace{\left(\sigma^2\right)^{-\underline{\alpha}-1} \exp\left(-\frac{\underline{\beta}}{\sigma^2}\right)}_{\text{prior for } \sigma^2}$$

- We will show that this is the posterior kernel of the normal inverse gamma (NIG) distribution.
- As before, we "complete the square."
- I'll derive it in class (you do not need to know this for an exam).

Posterior distribution

- This is the posterior kernel of a NIG distribution

$$p\left(\mu,\sigma^2|\mathbf{y}\right) \propto \underbrace{\left(\frac{1}{\sigma^2}\right)\exp\left(-\frac{1}{2\overline{\kappa}\sigma^2}\left(\mu-\overline{\mu}\right)^2\right)}_{\text{posterior for } \mu|\sigma^2}\underbrace{\left(\sigma^2\right)^{-\underline{\alpha}-1}\exp\left(-\frac{\underline{\beta}}{\sigma^2}\right)}_{\text{posterior for } \sigma^2}$$

- We find N $\left(\overline{\mu},\overline{\kappa}\sigma^2\right)$ Inv Gamma $\left(\overline{\alpha},\overline{\delta}\right)$

$$\begin{aligned}
\overline{\kappa}^{-1} &= \underline{\kappa}^{-1}+n \\
\overline{\mu} &= \overline{\kappa}\left(\underline{\kappa}^{-1}\underline{\mu}+\sum_{i=1}^{n}y_i\right) \\
\overline{\alpha} &= \underline{\alpha}+\frac{n}{2} \\
\overline{\delta} &= \underline{\delta}+\frac{1}{2}\left[\sum_{i=1}^{n}y_i^2+\underline{\mu}^2\underline{\kappa}^{-1}-\overline{\mu}^2\overline{\kappa}^{-1}\right]
\end{aligned}$$

Posterior mean of μ

- The posterior is

$$p\left(\mu|\mathbf{y},\sigma^2\right) \quad = \quad \mathsf{N}\left(\overline{\mu},\overline{\kappa}\sigma^2\right)$$

with posterior mean

$$\mathbb{E}\left(\mu|\mathbf{y}\right) \quad = \quad \overline{\kappa}\left(\underline{\kappa}^{-1}\underline{\mu}+\sum_{i=1}^{n}y_i\right)$$

- We can rearrange the posterior mean as

$$\mathbb{E}\left(\mu|\mathbf{y}\right) \quad = \quad \frac{\underline{\kappa}^{-1}}{\underline{\kappa}^{-1}+n}\underbrace{\underline{\mu}}_{\text{prior mean}}+\frac{n}{\underline{\kappa}^{-1}+n}\times\underbrace{\frac{1}{n}\sum_{i=1}^{n}y_i}_{\text{M.L.E.}}$$

- The posterior mean is a weighted average of the prior mean and maximum likelihood estimator.

Posterior variance and standard deviation

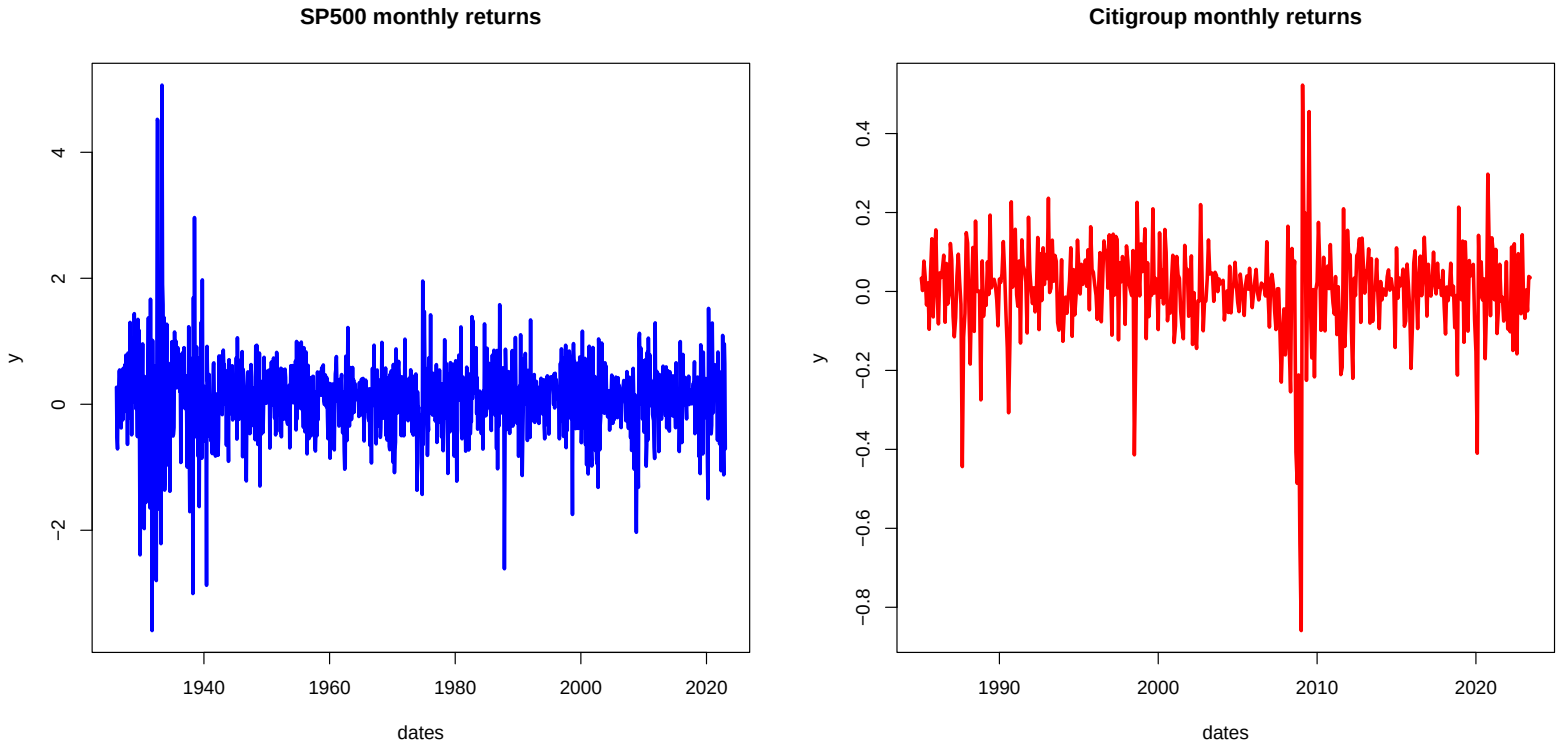
- The posterior (conditional) variance is

$$\mathbb{V}\left(\mu|\mathbf{y},\sigma^2\right) = \overline{\kappa}\sigma^2 = \frac{\sigma^2}{\underline{\kappa}^{-1}+n} = \frac{n}{\underline{\kappa}^{-1}+n} \times \left(\frac{\sigma^2}{n}\right)$$

- Remember: the frequentist standard error is $\sqrt{\frac{\sigma^2}{n}}$.

- In general, the Bayesian estimator will have smaller variance as long as $\underline{\kappa} > 0$.

Ex: S&P500 and Citi monthly returns



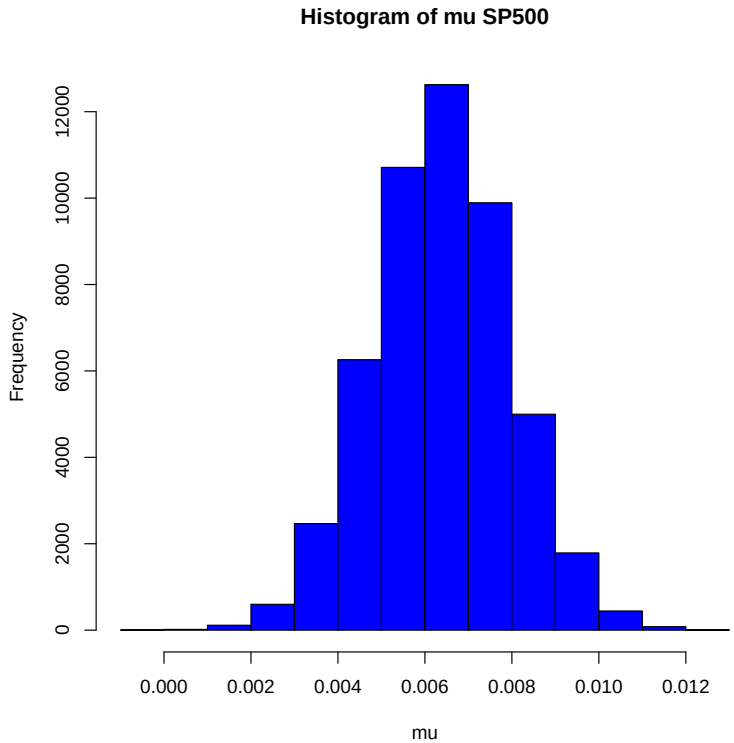
- Left: monthly SP500 returns, $n = 1164$, 1/1926 to 12/2022
- Right: Citigroup monthly returns, $n = 461$, 2/1985 to 7/2023

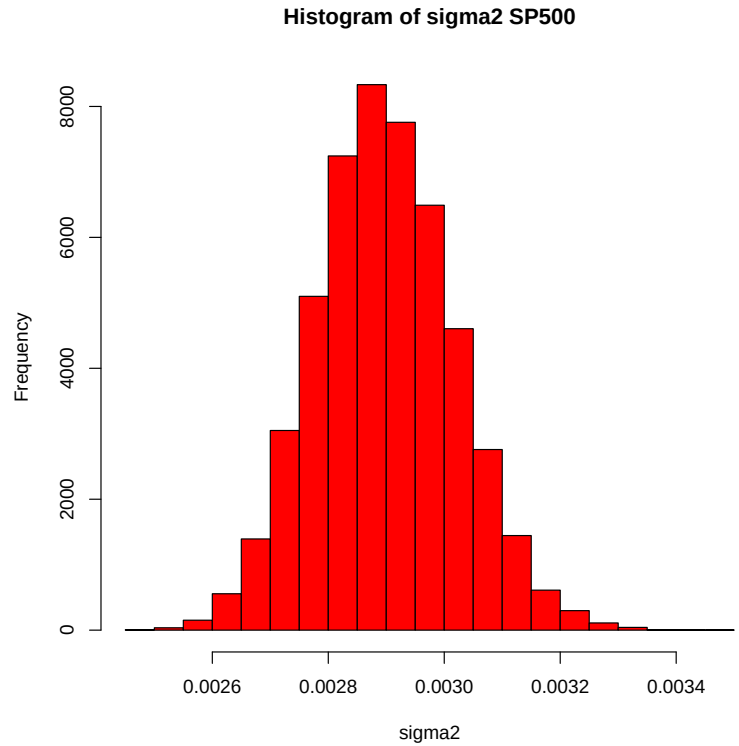
Comments

- I did not annualize the data because now we assume the variance σ^2 is unknown.
- I chose the prior hyperparameters as

$$\begin{aligned} \underline{\kappa} &= 0.25 \\ \underline{\mu} &= \frac{0.06}{12} \\ \underline{\alpha} &= 2.1 \\ \underline{\delta} &= 0.001 \end{aligned}$$

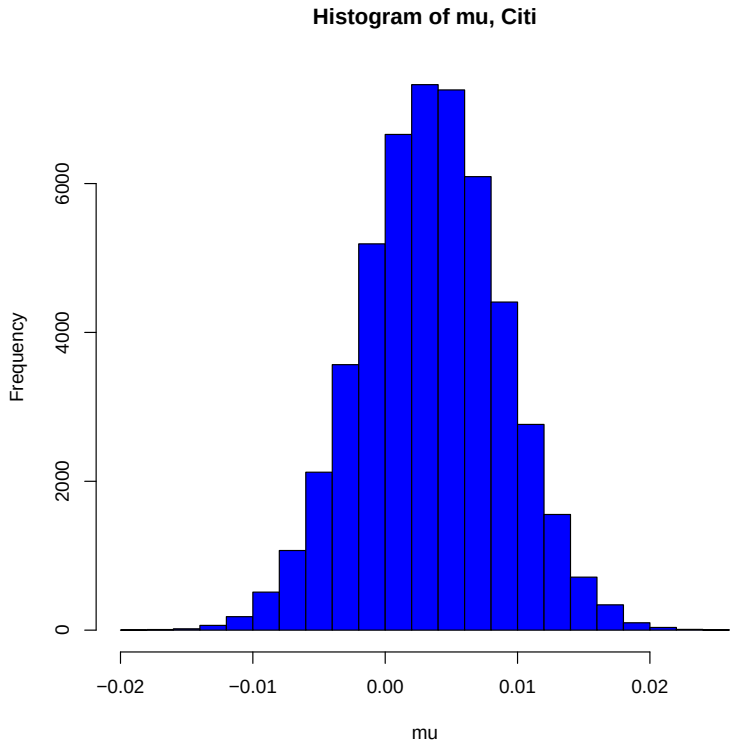
Ex: S&P500 monthly returns

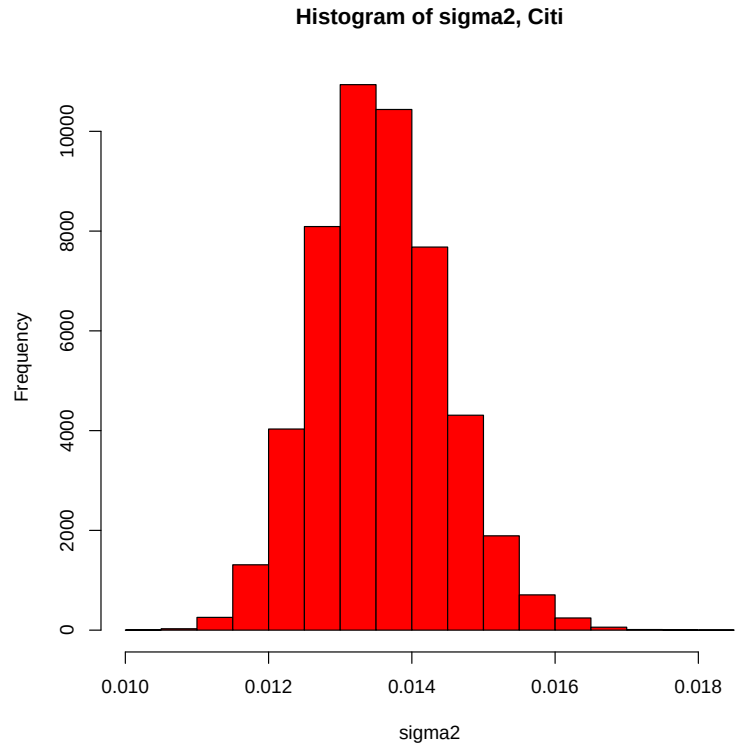




- $\widehat{\mu}_{mle} = 0.0063$, $\widehat{\sigma}^2_{mle} = 0.00290$, and $S.E.\left(\widehat{\mu}_{mle}\right) = 0.00158$
- Posterior: $\mathbb{E}\left(\mu|\mathbf{y}\right) = 0.0063$ and $S.D.\left(\mu|\mathbf{y}\right) = 0.00157$
- Posterior: $\mathbb{E}\left(\sigma^2|\mathbf{y}\right) = 0.00289$

Ex: Citi monthly returns





- $\widehat{\mu}_{mle} = 0.00348$, $\widehat{\sigma}^2_{mle} = 0.0136$, and $S.E.\left(\widehat{\mu}_{mle}\right) = 0.0054$
- Posterior: $\mathbb{E}\left(\mu|\mathbf{y}\right) = 0.0035$ and $S.D.\left(\mu|\mathbf{y}\right) = 0.0053$
- Posterior: $\mathbb{E}\left(\sigma^2|\mathbf{y}\right) = 0.0135$

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Posterior predictive distribution

Diffuse priors

- Diffuse priors are what we get if we want to make the prior variance go to infinity.
- For the NIG distribution, the diffuse prior is in the limit when $\underline{\kappa}^{-1} = 0$, $\underline{\alpha} = 0$, and $\underline{\delta} = 0$, i.e. the variances become infinite.
- The prior becomes

$$p\left(\mu, \sigma^2\right) \quad \propto \quad \frac{1}{\sigma^2}$$

- This is not a **proper** "probability density function" because it does not integrate to 1.

Posterior under a diffuse prior

- Under the diffuse prior, the posterior is **proper**.
- It is still a normal inverse gamma distribution.
- We find

$$\begin{aligned} p\left(\mu|\mathbf{y},\sigma^2\right) &= \mathrm{N}\left(\overline{\mu},\overline{\kappa}\sigma^2\right)\mathrm{Inv\ Gamma}\left(\overline{\alpha},\overline{\beta}\right) \\ \overline{\kappa}^{-1} &= n \\ \overline{\mu} &= \frac{1}{n}\sum_{i=1}^{n} y_i \\ \overline{\alpha} &= 1+\frac{n}{2} \\ \overline{\delta} &= \frac{1}{2}\left[\sum_{i=1}^{n} y_i^2 - \overline{\mu}^2\overline{\kappa}^{-1}\right] \end{aligned}$$

- This is equivalent to the MLE.

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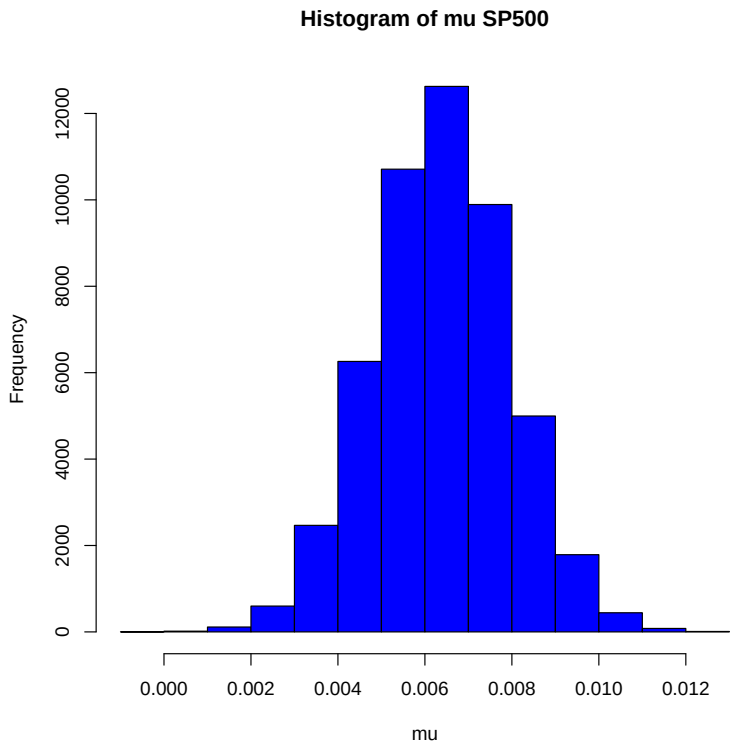
Marginal likelihood

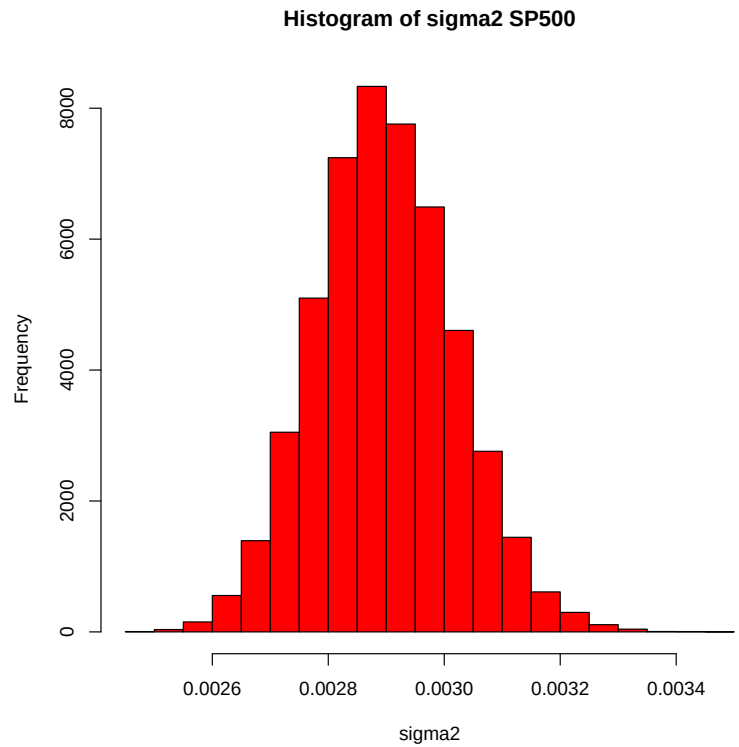
Posterior predictive distribution

Highest posterior density intervals

- I use the `R` package "`HDInterval`"
- Remember that Highest Posterior Density intervals (HPDI) are not "symmetric" if the distribution is asymmetric.

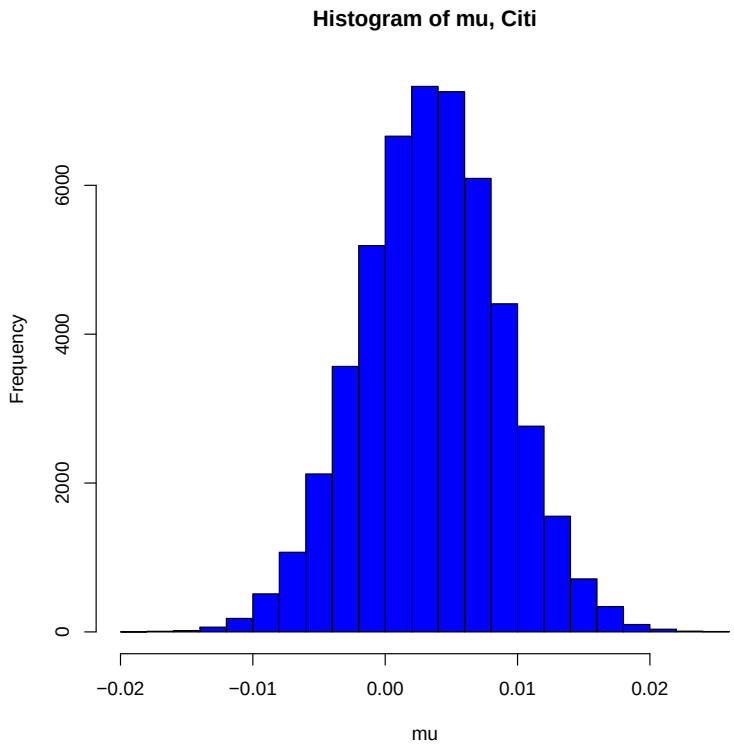
Ex: S&P500 monthly returns

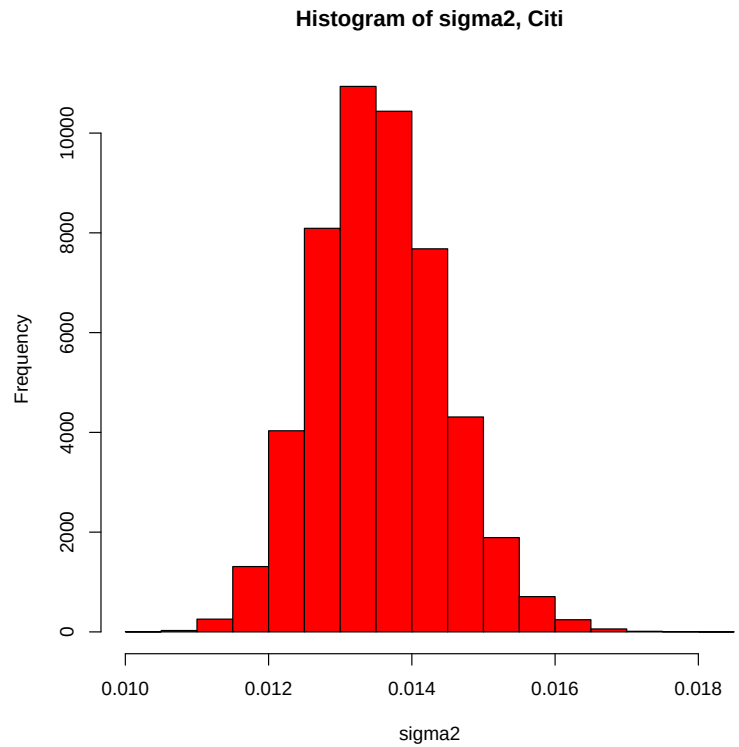




- 95% HPDI for μ: (0.0033,0.0095)
- 95% HPDI for σ^2: (0.0027, 0.00313)
- Frequentist 95% CI for μ: (0.0032,0.0095)

Ex: Citi monthly returns





- 95% HPDI for μ: (-0.0067, 0.0141)
- 95% HPDI for σ^2: (0.0118, 0.0153)
- Frequentist 95% CI for μ: (-0.0074,0.0143)

R code

```
# load the library
library(HDInterval)

# draw from the posterior
iNumDraws = 50000
post_draws = matrix(0,2,iNumDraws)
for(i in 1:iNumDraws)
{
sigma2_draw = 1/rgamma(1,shape=alpha_post,scale=(1/delta_post))
mu_draw = rnorm(1,mu_post,sqrt(kappa_post*sigma2_draw))
post_draws[1,i] = mu_draw
post_draws[2,i] = sigma2_draw
}

# Calculate the posterior via simulation
mean(post_draws[1,])
sd(post_draws[1,])
hdi(post_draws[1,],credMass=0.95)
```

Outline

Prior: normal distribution

Gaussian likelihood

Maximum likelihood estimation

Posterior distribution: unknown mean and known variance

Bias variance trade-off

Prior: normal inverse gamma distribution

Posterior distribution: unknown mean and variance

Diffuse priors

Highest posterior density intervals

Marginal likelihood

Posterior predictive distribution

Marginal likelihood

- The **marginal likelihood** is the normalizing constant in Bayes rule.

$$p(\mathbf{y}) = \int_0^\infty \int_{-\infty}^\infty p\left(\mathbf{y}|\mu,\sigma^2\right) p\left(\mu,\sigma^2\right) d\mu d\sigma^2$$

- For the normal model with NIG prior distribution, we can calculate this integral analytically.

$$p(\mathbf{y}) = \left(\frac{1}{2\pi}\right)^{\frac{n}{2}} \left(\frac{\overline{\kappa}}{\underline{\kappa}}\right)^{\frac{1}{2}} \frac{\Gamma\left(\overline{\alpha}\right)}{\Gamma\left(\underline{\alpha}\right)} \underline{\delta}^{\underline{\alpha}} \left(\overline{\delta}\right)^{-\frac{2\underline{\alpha}+n}{2}}$$

This happens to be the p.d.f. of a Student's t distribution.

Marginal likelihood

- The **marginal likelihood** is the normalizing constant in Bayes rule.
- Right now, we do not have any use for this.
- The marginal likelihood $p(\mathbf{y})$ is used in Bayesian hypothesis testing when you compare two models.
  - I'll show you examples in the future.

Outline

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Posterior predictive distribution

Posterior predictive distribution

- Suppose we have calculated the posterior $p\left(\mu, \sigma^2 | \mathbf{y}\right)$.
- And, we want to predict a new value $\widetilde{y}$ out of sample. The likelihood for the new data point is

$$\widetilde{y} \sim \mathrm{N}\left(\mu, \sigma^2\right)$$

- Remember, we have information about the parameters μ and σ^2 from our posterior $p\left(\mu, \sigma^2 | \mathbf{y}\right)$.
- What is the **posterior predictive distribution** for this model?

Posterior predictive distribution

- The **posterior predictive distribution** is

$$p\left(\widetilde{y}|\mathbf{y}\right) = \int_0^\infty \int_{-\infty}^\infty p\left(\widetilde{y}|\mu,\sigma^2\right) p\left(\mu,\sigma^2|\mathbf{y}\right) d\mu d\sigma^2$$

- It integrates out the uncertainty in the unknown parameter π by taking a weighted average over the posterior.
- For the normal model with NIG prior, the integral can be calculated by pencil and paper (next slide).
- But, it is much easier to calculate this by Monte Carlo simulation.

Posterior predictive distribution

- The **posterior predictive distribution** is

$$\begin{aligned} p\left(\widetilde{y}|\mathbf{y}\right) &= \int_0^\infty \int_{-\infty}^\infty p\left(\widetilde{y}|\mu,\sigma^2\right) p\left(\mu,\sigma^2|\mathbf{y}\right) d\mu d\sigma^2 \\ &= \left(\frac{1}{\sqrt{2\pi}}\right) \frac{\Gamma\left(\widetilde{a}\right)}{\Gamma\left(\overline{a}\right)} \overline{b}^{\overline{a}} \widetilde{b}^{-\widetilde{a}} \overline{\kappa}^{-\frac{1}{2}} \widetilde{\kappa}^{\frac{1}{2}} \end{aligned}$$

- The parameters are

$$\begin{aligned} \widetilde{\kappa}^{-1} &= 1 + \overline{\kappa}^{-1} \\ \widetilde{\mu} &= \widetilde{\kappa}\left[\overline{\kappa}^{-1}\overline{\mu} + \tilde{y}\right] \\ \widetilde{a} &= \overline{a} + \frac{1}{2} \\ \widetilde{b} &= \overline{b} + \frac{1}{2}\left[\tilde{y}^2 + \overline{\mu}^2\overline{\kappa}^{-1} - \tilde{\mu}^2\tilde{\kappa}^{-1}\right] \end{aligned}$$

R code

- Suppose we have already calculated $\overline{\alpha}, \overline{\delta}, \overline{\mu}, \overline{\kappa}$.

```
iNumDraws = 50000
y_draws = matrix(0,1,iNumDraws)
for(i in 1:iNumDraws)
{
sigma2_draw =
1/rgamma(1,shape=alpha_post,scale=(1/delta_post))
mu_draw = rnorm(1,mu_post,sqrt(kappa_post*sigma2_draw))
y_draw = rnorm(1,mu_draw,sqrt(sigma2_draw))
y_draws[1,i] = y_draw
}

# We can calculate the predictive via simulation
mean(y_draws[1,])
sd(y_draws[1,])
hdi(y_draws[1,],credMass=0.95)
```